



University of
Sheffield

School of
Computer
Science

COM6115 Text Processing

Assessment: Sentiment Analysis

Quick Summary

To better understand the strengths and limitations of Bayesian text classification, in this assignment, you are going to investigate Sentiment Analysis using two sentiment datasets you will be provided. You will also be provided with a Python script that implements Naïve Bayes. You will need to write a report (**no more than 1500 words**) to describe your results and findings.

Note: This assessment accounts for 30% of your total mark for the course. Your report may be submitted for a plagiarism check (e.g., Turnitin). For any clarification on this assessment, please use the Discussion Board on Blackboard.

Assessment Tasks

STEP 1

Download the data from Blackboard. This contains the following:

1. A dataset with snippets of movie reviews from the Rotten Tomatoes website (one text file for positive reviews and one text file for negative reviews):
 - a. `rt-polarity.pos`
 - b. `rt-polarity.neg`
2. A smaller dataset with snippets of reviews for Nokia phones (again, 2 files):
 - a. `nokia-pos.txt`
 - b. `nokia-neg.txt`
3. A sentiment dictionary of positive and negative sentiment words:
 - a. `negative-words.txt` contains 4783 negative-sentiment words
 - b. `positive-words.txt` contains 2006 positive-sentiment words
4. A Python script called `Sentiment.py` (you will need Python 3 to run it).

This includes: an implementation of Naïve Bayes, a knowledge-based classifier using the sentiment dictionary, and some helper functions.

STEP 2

Run Naïve Bayes on Rotten Tomatoes Data

The code splits the Rotten Tomatoes Data into a training and test set in `readFiles()`, then builds the $p(\text{word}|\text{sentiment})$ model on the training data in `trainBayes()`, and finally applies Naïve Bayes to the test data in `testBayes()`.

1. Observe the data in the positive and negative word dictionaries (`positive-words.txt` and `negative-words.txt`). Make any necessary changes in the code lines 24-28 to store the words in lists. Explain your decisions in the report.

[3 pt]

2. Write a function which will print out Accuracy, Precision, Recall and F-measure for the test data.

[5 pt]

3. Run the code and report the classification results.

[5 pt]

STEP 3

Run Naïve Bayes on Nokia Data

1. In the Python script, towards the end of the file (lines 276 and 278), uncomment out the other two calls to `testBayes()`. These run Naïve Bayes on the training data and on Nokia product reviews.
2. What do you observe? Why are the results so different?

[10 pt]

STEP 4

What is being learnt by the model?

1. Which are the most useful words for predicting sentiment?
The code you have downloaded contains another function `mostUseful()` that prints the most useful words for deciding sentiment. Uncomment the call to `mostUseful(pWordPos, pWordNeg, pWord, 100)` at the bottom of the program, and run the code again. This prints the words with the highest predictive value. Add these words (for both positive and negative sentiment) to your report. You can add them in an Appendix (these do not count to the total word count).

[5 pt]

2. Are the words selected by the model good sentiment terms?
How many of them are in the sentiment dictionary?

[5 pt]

STEP 5

How does a rule-based system compare to Naïve Bayes?

1. Add some code for the function `testDictionary()` which will print out Accuracy, Precision, Recall and F-measure for the test data.

[2 pt]

Uncomment out the three lines towards the end of the program that call the function `testDictionary()` and run the program again. All this code does is add up the number of negative and positive words present in a review and predict the larger class.

2. How does the dictionary-based approach compare to Naïve Bayes on the two domains? What conclusions do you draw about statistical and rule-based approaches from your observations?

[5 pt]

3. Write a new function to improve the rule-based system, e.g., to take into account negation, diminisher rules, etc and justify your decisions. Run the program again and analyse the results on both datasets.

[25 pt]

STEP 6

Error Analysis

1. Comment out all but one of the `testBayes` and `testDictionary` calls (one for each, two in total).
2. At the top of the program, set `PRINT_ERRORS=1`. Make any necessary changes to the definition of `testDictionary` so that errors are printed when called, similarly to `testBayes`.
3. Run the program again for each of the test calls, and it will print out the mistakes made. List the mistakes in the report.

[5pt]

4. Please explain why the model is making mistakes (e.g., analyse the errors and report any patterns or generalisations for each of the two test calls).

[15 pt]

Marking Criteria

Along with your code, you should submit a report to describe your results and findings by following the tasks detailed in the 6 steps above. You should also submit any additional files you may have used for implementing the improved rule-based system.

Here is a summary of the marking criteria:

1. Quality of the report, including structure and clarity. No more than 1500 words. **[15 pt]**
2. Step 2 **[13 pt]**
3. Step 3 **[10 pt]**
4. Step 4 **[10 pt]**
5. Step 5 **[32 pt]**
6. Step 6 **[20 pt]**

Submission Guidelines

You should submit the following via Blackboard by the deadline:

- a PDF version of your report
- your code
- any additional files you may have used for implementing the improved rule-based system.